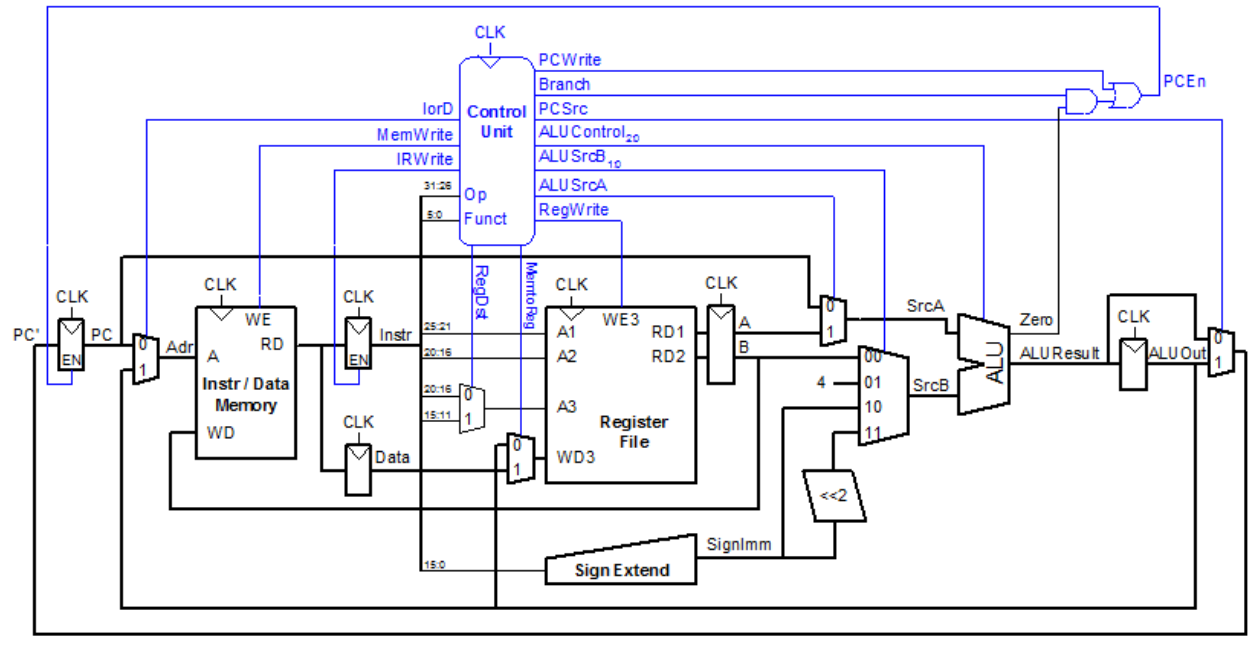


CS2206: Computer architecture

Lab 09: Study of MIPS multi- Cycle Processor implementation



Task 1: Study the given multi-cycle design and identify errors. Convert the following C code to MIPS assembly, test the code in MIPS In the multicycle cycle implementation of the processor. Assume appropriate memory address for storage of the array.

```
for(i = 0; i < 100; i++)
```

```
  A[i] = i;
```

(15 points)

Task 2: Write ASCII values of your name full name and roll number to the memory using appropriate instruction, and write program to find the sum of those values using the MIPS given MIPS multi-cycle design

(15 points)

Task 3: Simulate the following snippet of code. Identify the functionality (assume suitable array if necessary) and Compute CPI for the given design and simulate

```
addi $s7, $0, 9

loop1: addi $s6, $0, 0

add $s5, $0, $0

loop2: lw $s0, 0($s6)

lw $s1, 4($s6)

slt $s2, $s0, $s1

beq $s2, $0, swap

j next

swap: add $t0, $0, $s0

add $s0, $0, $s1

add $s1, $0, $t0

next: sw $s0, 0($s6)

sw $s1, 4($s6)

addi $s5, $s5, 1

addi $s6, $s6, 4

beq $s5, $s7, cont

j loop2

cont: subi $s7, $s7, 1

beq $s7, $0, stop

j loop1

stop: exit
```

(30 points)

Your CPI computation in record;

Task4:

Implement a new instruction in the given multi-cycle design and test using appropriate program. Add the new design as a library component. Test with appropriate test program

(40 points)

Submission : Submit your design and source code to (in a single file- with your roll name as file name to

<https://u.pcloud.com/#/puplink?code=eX1XZeYpOFUlltQhg8FrgbXiEkrAXq2V>